



Question Bank with Answers

Chapter 2 — Cybersecurity

Book-style question formats

PROGRAMMING AND ARTIFICIAL INTELLIGENCE | SEMESTER 1

81 Questions Across 4 Sections

Fill in the blank • True/False • Matching • Multiple choice • Short answer • Analysis

QUESTION BANK

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Every question is reproduced exactly as written, grouped by lesson. Each question is tagged by format (fill in the blank, true/false, matching, multiple choice, short answer, analysis, or extended response) with its answer shown directly beneath in a green box.

2-1 • CHAPTER 2 QUESTION BANK

2-1

Cryptographic Technologies and Authentication

TRUE / FALSE

Q1A

HTTPS combines public-key cryptography and common-key cryptography. (True or False?)

ANSWER

True — Public-key cryptography is used for safe key sharing; common-key cryptography is used for data communication.

TRUE / FALSE

Q1B

HTTPS uses public-key cryptography to exchange all data after the common key is shared. (True or False?)

ANSWER

False — Common-key cryptography is used for data communication because it is faster.

TRUE / FALSE

Q1C

Multi-factor authentication combines two or more different authentication factors. (True or False?)

ANSWER

True — The factors are knowledge, possession, and biometric.

TRUE / FALSE

Q1D

Using two different passwords is an example of multi-factor authentication. (True or False?)

ANSWER

False — Both passwords are knowledge factors.

SHORT ANSWER

Q2.1

In HTTPS, which cryptographic method is used so the browser and server can safely share a common key?

ANSWER

Public-key cryptography.

SHORT ANSWER

Q2.2

Which cryptographic method is used for data communication after the common key is shared?

ANSWER

Common-key cryptography (symmetric-key cryptography).

SHORT ANSWER

Q2.3

Why are the two cryptographic methods combined?

ANSWER

Public-key cryptography shares the key safely, while common-key cryptography exchanges large amounts of data quickly. This provides both security and speed.

MATCHING

Q3

Match each description with the most appropriate choice. Choices: A Encryption • B Digital signature • C Digital certificate • D Multi-factor authentication

- a. Prevent communication content from being read by third parties.
- b. Detect whether data has been altered during communication.
- c. Verify that the communication partner is authentic.
- d. Prevent unauthorized login even if a password is leaked.

ANSWER

- a → A
- b → B
- c → C
- d → D

MATCHING

Q4

Match each description with the most appropriate choice. Choices: A Knowledge + possession • B Knowledge + biometric • C Biometric + possession

- a. Password + one-time password
- b. Password + fingerprint authentication
- c. Password + SMS authentication
- d. Fingerprint authentication + smart card

ANSWER

- a → A
- b → B
- c → A
- d → C

MULTIPLE CHOICE

Q5

Which is NOT an appropriate description of security technologies?

- A. Encryption can prevent communication content from being read by third parties.
- B. A digital signature is mainly used to improve communication speed.
- C. A digital certificate helps verify that the communication partner is authentic.
- D. Combining several security technologies can address different threats.

ANSWER

B — Improving communication speed is not the purpose of a digital signature.

FILL IN THE BLANK

Q6

The web server sends a (a) and a public key to the browser. The browser verifies the server through (b). Next, the browser creates a (c) for communication and protects it using the server's public key. Data communication then uses (c).

ANSWER

- (a) digital certificate
- (b) certificate authority (CA)
- (c) common key

MULTIPLE CHOICE

Q7

Which is NOT an appropriate example of multi-factor authentication?

- A. Password + smartphone notification
- B. Password + secret question
- C. Fingerprint authentication + smart card
- D. Password + SMS authentication

ANSWER

B — A password and a secret question are both knowledge factors.

SHORT ANSWER

Q8.1

Name the three authentication factors.

ANSWER

Knowledge, possession, and biometric.

SHORT ANSWER

Q8.2

What does a digital certificate verify?

ANSWER

It verifies that the communication partner, such as a website, is authentic.

SHORT ANSWER

Q8.3

What does a digital signature help detect?

ANSWER

Data tampering and impersonation; it also provides non-repudiation.

6-MARK

Q9 6 marks

Explain why HTTPS uses public-key cryptography to share the key but common-key cryptography to exchange the data.

ANSWER

- Public-key cryptography is used to share the common key safely.
- The browser and server use the public/private key process during key sharing.
- After the common key is established, common-key cryptography is used for the data.
- Common-key cryptography is faster for large amounts of data.
- Using public-key cryptography for all data communication would be inefficient.
- Therefore, combining both methods provides security and communication speed.

THINK AS AN ENGINEER**Q10**

A school portal lets students check grades and pay fees. Which is better: password only or password plus a one-time password sent to the phone? Give evidence.

ANSWER

Password + one-time password is the stronger choice because it combines knowledge and possession. If the password leaks, the second factor can still prevent unauthorized access. A student without a smartphone may need an alternative.

NEW CONTEXT**Q11**

A small online shop wants to secure customer login and payment. Name two security technologies, say where each helps, and identify one threat that could still remain.

ANSWER

For example, HTTPS/encryption protects communication content and MFA protects account login. A digital certificate can verify the genuine site. A remaining risk could be a compromised user device or account.

2-2 • CHAPTER 2 QUESTION BANK

2-2

Network Security Design

TRUE / FALSE

Q1A

A firewall allows or denies communication based on rules set in advance. (True or False?)

ANSWER

True — This is the role of a firewall at the network entry/exit point.

TRUE / FALSE

Q1B

Using a VPN can reduce the risk of eavesdropping on public Wi-Fi. (True or False?)

ANSWER

True — VPN communication is encrypted.

TRUE / FALSE

Q1C

A DMZ exposes all servers directly to the outside. (True or False?)

ANSWER

False — A DMZ separates public-facing servers from the internal network.

TRUE / FALSE

Q1D

Defense in depth means layering multiple security measures. (True or False?)

ANSWER

True — If one layer is breached, another layer can still protect the system.

SHORT ANSWER

Q2.1

What is the term for a system at a network entry/exit point that allows or denies communication according to rules?

ANSWER

Firewall.

SHORT ANSWER

Q2.2

What is the three-letter abbreviation for an encrypted virtual private line over the Internet?

ANSWER

VPN — Virtual Private Network.

SHORT ANSWER

Q2.3

What is the three-letter abbreviation for the area where public-facing servers are separated from the internal network?

ANSWER

DMZ — Demilitarized Zone.

SHORT ANSWER

Q2.4

What is the concept of layering multiple security measures?

ANSWER

Defense in depth.

SHORT ANSWER

Q2.5

What concept means that no communication is trusted automatically and every access is verified?

ANSWER

Zero trust.

MATCHING

Q3

Match each description with the most appropriate choice. Choices: A Firewall • B VPN • C DMZ • D Defense in depth

- a. Creates an encrypted virtual private line over the Internet for safe remote connection.
- b. Places public-facing servers separately from the internal network.
- c. Allows or denies communication based on preset rules.
- d. Layers multiple security measures.

ANSWER

- a → B
- b → C
- c → A
- d → D

MULTIPLE CHOICE

Q4

Which is NOT an appropriate use of a VPN?

- A. Connecting safely from home to an organization's network.
- B. Connecting offices in different locations safely.
- C. Automatically deleting files infected with viruses.
- D. Reducing eavesdropping risk on public Wi-Fi through encrypted communication.

ANSWER

C — Deleting infected files is not a VPN function.

MATCHING

Q5

Match each description with the most appropriate choice. Choices: A Network entry • B Communication path • C Server placement • D Endpoints

- a. Blocking unauthorized communication with a firewall.
- b. Encrypting communication with a VPN.
- c. Installing antivirus software and keeping the operating system updated.
- d. Separating public-facing servers using a DMZ.

ANSWER

- a → A
- b → B
- c → D
- d → C

MULTIPLE CHOICE

Q6

What is the most appropriate reason for providing a DMZ?

- A. To improve communication speed.
- B. To expose all servers to the outside.
- C. To reduce the chance that an attack on a public-facing server reaches the internal network.
- D. To automatically detect and remove viruses.

ANSWER

C — A DMZ separates public-facing servers from the internal network.

MULTIPLE CHOICE

Q7

Which statement most appropriately describes zero trust?

- A. Access from inside is safe, so verification is unnecessary.
- B. A firewall alone is sufficient.
- C. No communication is trusted automatically and every access is verified.
- D. Only outside access needs monitoring.

ANSWER

C — Zero trust verifies every access, including internal access.

SHORT ANSWER

Q8.1

Give two main uses of a VPN.

ANSWER

Safe remote connection to an organization's network and safe connection between offices/branches.

SHORT ANSWER

Q8.2

Why is a public-facing web server placed in a DMZ?

ANSWER

To separate it from the internal network so an attack on the public server is less likely to reach internal data.

SHORT ANSWER

Q8.3

Why is a firewall alone not enough?

ANSWER

Other layers can still limit damage if the firewall or another measure is breached.

SHORT ANSWER

Q8.4

Why has the traditional perimeter model become less sufficient?

ANSWER

Cloud services and remote work have blurred the boundary between inside and outside.

6-MARK

Q9 6 marks

Explain why defense in depth protects an organization better than relying on a firewall alone.

ANSWER

- A firewall controls communication entering or leaving the network.
- A VPN protects communication paths, especially remote connections.
- A DMZ separates public-facing servers from the internal network.
- Endpoint protection such as antivirus and updates protects individual devices.
- If one measure is breached, another layer can still limit damage.
- Therefore, layered security is stronger than a single barrier.

THINK AS AN ENGINEER

Q10

A company has a public website, staff working from home, and confidential customer data. Recommend a layered design and give two reasons it is safer than a firewall alone.

ANSWER

Use a firewall at network entry, a VPN for remote staff, a DMZ for the public website, and endpoint protection for devices. Zero trust should verify access. Different layers protect different paths, and if one layer fails the others can still limit damage.

NEW CONTEXT

Q11

A hospital stores patient records in a cloud service and allows doctors to access them from home. Using zero trust and defense in depth, suggest two measures, explain where each helps, and identify one remaining risk.

ANSWER

Require identity/permission verification for every access and use encrypted remote communication such as a VPN. Endpoint protection can also protect devices. A remaining risk could be a compromised account or device.

2-3 • CHAPTER 2 QUESTION BANK

2-3

Incident Response and Risk Management

TRUE / FALSE

Q1A

A security incident is a situation in which actual damage occurs because of an information security problem. (True or False?)

ANSWER

True — This is the chapter's definition.

TRUE / FALSE

Q1B

Containment is performed before eradication so the spread of damage can be limited. (True or False?)

ANSWER

True — Containment limits the scope before the cause is removed.

TRUE / FALSE

Q1C

Risk size can be determined by impact alone. (True or False?)

ANSWER

False — Risk size = impact × likelihood.

TRUE / FALSE

Q1D

Prevention of recurrence is the final stage of the six-stage response. (True or False?)

ANSWER

True — The final stage analyzes the cause and prevents recurrence.

SHORT ANSWER

Q2.1

What is a security incident?

ANSWER

A situation in which actual damage occurs due to an information security problem.

SHORT ANSWER

Q2.2

Name the six stages of incident response in order.

ANSWER

Preparation → Detection → Containment → Eradication → Recovery → Prevention of recurrence.

SHORT ANSWER

Q2.3

What two factors determine risk size?

ANSWER

Impact and likelihood.

SHORT ANSWER

Q2.4

What does impact mean?

ANSWER

How much damage occurs if the risk actually happens.

SHORT ANSWER

Q2.5

What does likelihood mean?

ANSWER

How probable it is that the risk will actually happen.

MATCHING

Q3

Match each description with the most appropriate choice. Choices: A Preparation • B Detection • C Containment • D Eradication • E Recovery • F Prevention of recurrence

- a. Establish response procedures in advance.
- b. Discover and confirm that an incident has occurred.
- c. Limit the scope of impact so damage does not spread.
- d. Remove the cause, such as malware.
- e. Return systems and services to normal.
- f. Analyze the cause and prevent recurrence.

ANSWER

- a → A
- b → B
- c → C
- d → D
- e → E
- f → F

MULTIPLE CHOICE

Q4

Which is NOT an appropriate example of a security incident?

- A. Data is encrypted by a ransomware infection.
- B. Customer information is leaked through unauthorized access.
- C. A normal software update is performed to add new features.
- D. A web service becomes unavailable because of an attack.

ANSWER

C — A normal software update is not a security incident.

MULTIPLE CHOICE

Q5

Which statement best describes risk assessment?

- A. Only large-impact risks need attention.
- B. Only high-likelihood risks need attention.
- C. Risks with both high impact and high likelihood should be addressed first.
- D. All risks must be addressed simultaneously.

ANSWER

C — Risk size is impact × likelihood, so high-high risks have the highest priority.

FILL IN THE BLANK

Q6

A situation in which actual damage occurs because of an information security problem is called (a). When it occurs, (b), which limits the spread of damage, is performed before (c), which removes the cause. After that, (d) returns systems to normal, followed by (e), which prevents recurrence.

ANSWER

- (a) security incident
- (b) containment
- (c) eradication
- (d) recovery
- (e) prevention of recurrence

SHORT ANSWER

Q7.1

Give three examples of security incidents.

ANSWER

Information leak, data tampering, and service outage.

SHORT ANSWER

Q7.2

Why must containment come before eradication?

ANSWER

Containment limits the spread of damage before the cause is removed.

SHORT ANSWER

Q7.3

Why should recovery not be performed before the cause is removed?

ANSWER

If the cause, such as malware, remains, the same incident may happen again.

SHORT ANSWER

Q7.4

What is the risk formula?

ANSWER

Risk size = Impact × Likelihood.

SHORT ANSWER

Q7.5

Which risks should be addressed first?

ANSWER

Risks with both high impact and high likelihood.

6-MARK

Q8 6 marks

Explain why the six stages of incident response should be carried out in order, using eradication and recovery as examples.

ANSWER

- Preparation and detection make the organization ready and confirm the incident.
- Containment limits the spread of damage.
- Eradication removes the cause, such as malware.
- Recovery returns systems and services to normal.
- Recovering before eradication can leave the cause in the system.
- Therefore, the ordered stages help limit damage and prevent the same incident from returning.

THINK AS AN ENGINEER

Q9

A school's student-record system is hit by ransomware: files are encrypted and staff cannot log in. The incident has just been detected. What stages should be carried out next?

ANSWER

After detection, carry out containment to limit spread, eradication to remove the ransomware/cause, recovery to restore clean systems/data, and prevention of recurrence to analyze the cause and prevent another attack.

RISK-PRIORITIZATION

Q10

A school faces two risks: another ransomware attack and a brief power cut. How should impact and likelihood be used to decide which to address first?

ANSWER

Estimate the impact and likelihood of each risk. The risk with the higher combination should receive higher priority. A high-impact, high-likelihood ransomware risk should be addressed before a lower risk.

NEW CONTEXT

Q11

An online shop has detected that customer data has been leaked. Using the six-stage response and risk assessment, name the next two stages and explain why. Then identify one risk to prioritize afterward.

ANSWER

After detection, containment should limit further leakage, followed by eradication to remove the cause. Then assess remaining risks using impact × likelihood and prioritize a high-impact, high-likelihood risk.

REFLECT & CHALLENGE

Q12.1

Which stage of incident response might be hardest under pressure? Give a reason.

ANSWER

Answers may vary. For example, containment can be difficult because decisions must be made quickly to stop damage spreading while the full cause may not yet be known.

REFLECT & CHALLENGE

Q12.2

Suggest one thing an organization can prepare in advance to make incident response faster.

ANSWER

Examples: a documented response procedure, assigned responsibilities, communication contacts, backup arrangements, or predefined isolation procedures.

REVIEW • CHAPTER 2 QUESTION BANK

Review**Chapter 2 — Mixed Revision****SHORT ANSWER****QA1**

What is HTTPS?

ANSWER

HTTP with TLS encryption added; it protects communication through encryption, tamper detection, and verification of the communication partner.

SHORT ANSWER**QA2**

What is the main role of public-key cryptography in HTTPS?

ANSWER

Safe key sharing.

SHORT ANSWER**QA3**

What is the main role of common-key cryptography?

ANSWER

Fast data communication after the common key is shared.

SHORT ANSWER**QA4**

What does MFA require?

ANSWER

Two or more different authentication factors.

SHORT ANSWER**QA5**

What does a firewall do?

ANSWER

Allows or denies communication at a network entry/exit point according to rules.

SHORT ANSWER**QA6**

What does a VPN provide?

ANSWER

An encrypted virtual private line/tunnel over the Internet.

SHORT ANSWER

QA7

What does a DMZ separate?

ANSWER

Public-facing servers from the internal network.

SHORT ANSWER

QA8

What is defense in depth?

ANSWER

Layering multiple security measures.

SHORT ANSWER

QA9

What is zero trust?

ANSWER

No communication is trusted automatically; every access is verified.

SHORT ANSWER

QA10

What is a security incident?

ANSWER

A situation in which actual damage occurs due to an information security problem.

SHORT ANSWER

QA11

What is containment?

ANSWER

Limiting the scope of impact so damage does not spread.

SHORT ANSWER

QA12

What is eradication?

ANSWER

Removing the cause of the incident.

SHORT ANSWER

QA13

What is recovery?

ANSWER

Returning systems and services to their normal state.

SHORT ANSWER

QA14

What is risk size?

ANSWER

Impact × likelihood.

MULTIPLE CHOICE

QB1

Which pair is correctly matched?

- A. Digital certificate — verifies the communication partner
- B. VPN — removes malware from an endpoint
- C. DMZ — exposes all servers directly
- D. MFA — uses only one factor

ANSWER

A — A digital certificate verifies the communication partner.

MULTIPLE CHOICE

QB2

Which sequence is correct?

- A. Detection → Preparation → Recovery → Containment
- B. Preparation → Detection → Containment → Eradication → Recovery → Prevention of recurrence
- C. Preparation → Recovery → Detection → Eradication
- D. Containment → Recovery → Detection → Prevention

ANSWER

B — This is the six-stage order given in Chapter 2.

MULTIPLE CHOICE

QB3

Which combination best represents defense in depth?

- A. Firewall only
- B. Password only
- C. Firewall + VPN + DMZ + endpoint protection
- D. DMZ only

ANSWER

C — Defense in depth uses multiple layers.

6-MARK

QC1 6 marks

A company wants to protect an online service. Explain how cryptographic technologies, authentication, network security design, incident response, and risk assessment can work together.

ANSWER

- HTTPS can use public-key cryptography for safe key sharing and common-key cryptography for fast data communication.
- Digital certificates can verify the genuine communication partner and MFA can reduce unauthorized login risk.
- Firewall, VPN, DMZ, and endpoint protection provide defense in depth.
- Zero trust verifies access instead of automatically trusting it.
- If an incident occurs, follow preparation, detection, containment, eradication, recovery, and prevention of recurrence.
- Prioritize risks using $\text{impact} \times \text{likelihood}$, addressing high-impact and high-likelihood risks first.